

Mamenchisaurus



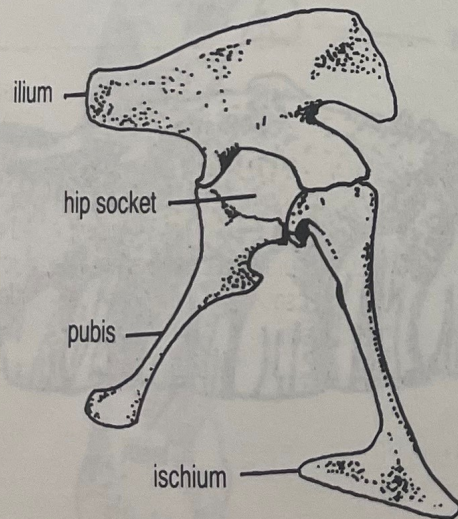
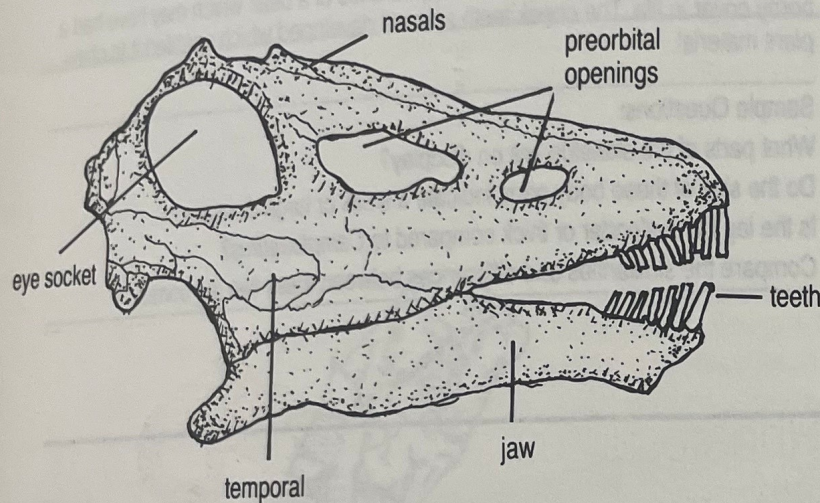
(ma-men-chi-sawr-us)

A quadrupedal plant-eating saurischian ('lizard-hipped') dinosaur (sauropod).

Meaning: 'Reptile from Mamenchì'.

On Display: Reconstruction of near-complete skeleton found in China (Sichuan 'Szechuan').

Size: 22 metres long.



For Discussion: features listed can be easily seen.

Skull: relatively small (with relatively small brain), fairly lightly built (large openings behind the eye sockets: *temporals* and two large openings in front of the eye: *preorbitals*); eye and nasal cavities high up on the skull (may indicate that it was at least partly aquatic, although may suggest it was terrestrial and had a special trunk like a tapir or elephant (with nasals high on the skull); had small peglike teeth in the front of small jaws (indicating a lack of powerful muscles; suggests it ate soft plants, eg. water plants, although some fossil teeth were well worn and suggest it browsed on tough plants, eg. pine needles).

Neck: longer than tail (11 metres – longest of any known sauropod); more than twenty hollowed vertebrae (lightened skeleton) with U-shaped notches (carrying cable-like ligament which provided support) (probably stretched it up to reach tall trees, eg. conifers, pines, firs, to eat food, or lowered it to eat softer plants).

Front Limbs: relatively short and thick with short thick foot bones; five toes (size may indicate bipedal ancestry); straight pillar-like (stood erect like elephant with the sole slightly raised).

Hind Limbs: relatively longer; also very thick and pillar-like (to support body); toes relatively short and thick with large claws (to prevent slipping).

Tail: very long, vertebrae near hipbones with large spines (for large, thick muscles), some fused (rigidity and provided more support); 7 metres long.

Hip Bones: similar to *Gallimimus*, under the hipbone (ilium) socket one bone points down and forward (pubis) and the other points down and backwards (ischium) like lizards.

Sample Questions:

Does it stand on two or four legs? On its whole foot or just its toes?

Is the tail longer or shorter than the neck?

Estimate the lengths of the tail and the neck.

How might the peg-like teeth have been used?